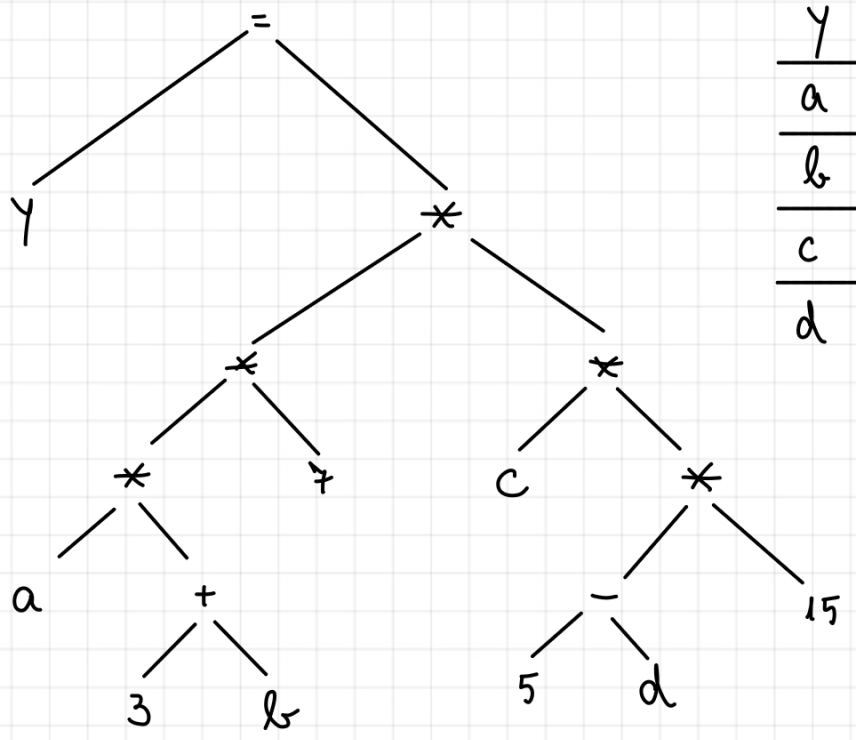
1.

Subject A

1. Using the assignment compilation example imagine the optimization for the expression y= (a* (3+b) *7) * (c* (5-d) *15). Write the followings: i) the intermediate code; ii) the optimized code; iii) the final code.

$y = (a * (3 + b) * 7) * (c * (5 - d) * 15)$



y	id1
a	id2
b	id3
c	id4
d	id5

i) temp 1 = inttoreal (3)

temp

2.

2. Consider the following code:

```
int tab[3];
void p2(int &x, int &y)
{
   x=y*y;
   y=x*x;
}
void p1(int &x, int &y)
{
   p2(y,x);
   x=y*y;
   y=x*x;
}
void main()
{
   tab[0]=3;
   tab[2]=4;
   p1(tab[0],tab[2]);
}
```

What are the values and how were they obtained for the array elements **tab** after the call to procedure **p1** if parameter transmission is made by reference.

	0	1	2
tab	3	0	4

p1 (tab[0], tab[2]) -> p2 (tab[2], tab[0])

-> tab[2] = tab[0] * tab[0] = 9

tab[0] = tab[2] * tab[2] = 81

tab[0] = tab[2] * tab[2] = 9 * 9 = 81

tab[2] = tab[0] * tab[0] = 81 * 81 = 6561

3.

3. Write a type in C containing an array of pointers which point to the type itself and a program that creates a 3 level tree having the degree of 2 using that type.

```
#include <stdio.h>
#include <stdlib.h>

typedef struct Node{
        int value;
        struct Node * child[2];
} Node;

Node * createNode(int value){

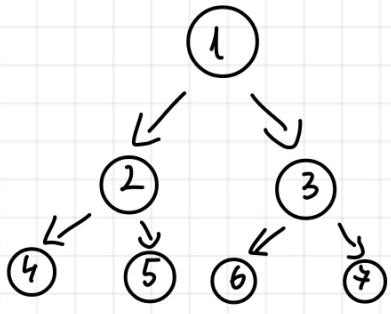
    Node * node = (Node*) malloc(sizeof(Node));
    node->value = value;
    node->child[0] = NULL;
    node-> child[1] = NULL;
    return node
}
int main(void){

    Node * n = createNode(1);

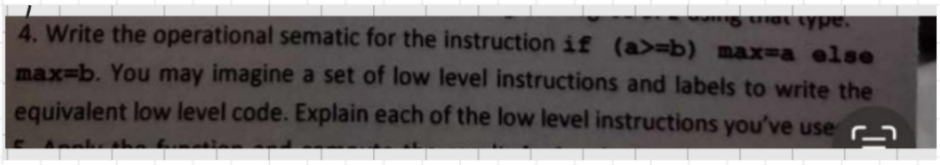
    n->child[0] = createNode(2);
    n->child[1] = createNode(3);

    n->child[0]->child[0] = createNode(4);
    n->child[0]->child[1] = createNode(5);
    n->child[1]->child[0] = createNode(6);
    n->child[1]->child[1] = createNode(7);

    return 0;
}
```



(4.)



4. Write the operational sematic for the instruction **if (a>=b) max=a else max=b**. You may imagine a set of low level instructions and labels to write the equivalent low level code. Explain each of the low level instructions you've use

```
if( a>=b){
        max=a
}
else{
        max=b
}
```

if (x>=y) then max := x
else max := y endif

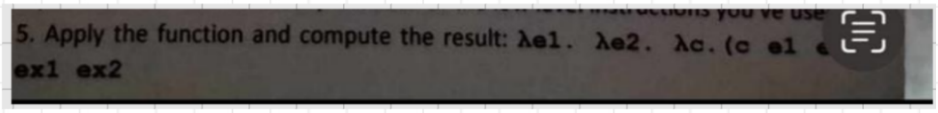
semantică operațională -> MV

```
if a < b goto ELSE
    max := a
ELSE :
    max := b
EXIT: HALT
```

parte citire a -> AX b -> BX

```
CMP AX, BX
JL ELSE
MOV CX, AX
JMP EMB
ELSE:
    MOV CX, BX
EMB : NOOP
```

(5.)


5. Apply the function and compute the result: λe1. λe2. λc. (c e1 e
ex1 ex2

$\lambda e1.\ \lambda e2.\ \lambda c.(c\ \ e1\ \ e2)\ \ ex1\ \ ex2$

$\lambda e2.\ \lambda c.(\ c\ \ ex1\ \ e2)\ \ \ ex2$

$\lambda c.\ (c\ \ \ ex1\ \ ex2)$